Problem

Title

Neon Number Verification

Topics

Number Mathematics, Square Calculation, Digit Extraction, Loops, Conditional Statements

Difficulty Level

Medium

Problem Detailed Statement (Scenario Based)

In a mathematical reasoning session, students are introduced to a special category of numbers called **Neon Numbers**.

A **Neon Number** is a number where the sum of digits of its square is equal to the number itself.

In other words:

1. First, calculate the square of the number.
2. Extract all digits of the square.
3. Add those digits.
4. If the sum equals the original number, it is called a Neon Number.

For example:

- 9
  ($9^2 = 81$)
  Sum of digits = 8 + 1 = 9 ✔
- 1
  ($1^2 = 1$)
  Sum of digits = 1 ✔

However:

- 4
  ($4^2 = 16$)
  Sum of digits = 1 + 6 = 7 ✖

Now, the trainer provides a positive integer **N**. Students must determine whether the given number is a Neon Number.

Students are required to:

- Compute the square of N.
- Extract digits using loop logic (% and / operators).
- Calculate the sum of digits.
- Compare the sum with the original number.
- Avoid converting the number into a string.
- Use proper looping and conditional structures.

This problem helps students strengthen:

- Digit manipulation logic
- Mathematical reasoning
- Loop-based processing
- Condition-based classification

Edge cases to consider:

- Single digit numbers
- Zero
- Large input values

Input Format

A single integer:

`N`

Constraints:

`0 ≤ N ≤ 10^6`

Output Format

- Print `"Neon Number"` if N is a Neon Number.
- Otherwise print `"Not Neon Number"`.

Examples

Example 1

Input:

```
9
```

Output:

```
Neon Number
```

Example 2

Input:

```
1
```

Output:

```
Neon Number
```

Example 3

Input:

```
4
```

Output:

```
Not Neon Number
```

Example 4

Input:

```
12
```

Output:

```
Not Neon Number
```
